

Pinzhe Zhao

Zach Zhao · Final-year Cognitive Science undergraduate · University of Edinburgh

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Research Interests

Computational cognitive science; human abstraction and learning; self-directed learning; flexible problem solving; human-inspired AI.

Education

Present **University of Edinburgh**
Cognitive Science MA (Hons); expected graduation June 2027.
Expected classification: First-Class Honours.
Coursework in computer science and cognitive science.

Research Experience

- Jun 2026–
present **Summer Research Intern** · University of Edinburgh
Computational Cognitive Science Lab · School of Informatics
Supervisor: Bonan Zhao
Funded by the Institute for Language, Cognition and Computation
- Investigating when people move beyond an existing representation to explore alternative ways of understanding and solving problems.
- Sep 2025–
present **Research Assistant** · University of Edinburgh
Computational Cognitive Science Lab · School of Informatics
Supervisors: Bonan Zhao and Marta Kryven
- Studying human abstraction learning: how people acquire and reuse structured knowledge to solve current and future problems.
- Jun–Aug 2025 **Visiting Student Research Assistant**
Shenzhen Medical Academy of Research and Translation (SMART)
Supervisor: Yuanyuan Yao
- Built an EEG/ECG pipeline for a mouse sleep study, covering signal cleaning, feature extraction, visualisation, and analysis by sleep state.
 - Developed methods to detect theta/delta surges and predict sinus pauses from neural, cardiac, and sleep-state features.
- Dec 2024–
Jun 2025 **Voluntary Research Assistant** · University of Sheffield
Cognitive Ability and Plasticity Lab · Supervisor: Claudia von Bastian
- Analysed pre-training, post-training, and follow-up data from a longitudinal processing-speed training study to examine individual differences in transfer.
 - Modelled transfer outcomes and estimated cognitive parameters from behavioural data.

Teaching

Sep 2026–
present **Marker (Teaching Support Provider)** · University of Edinburgh
School of Informatics

Volunteer Mathematics Teacher
Alba Cathay Chinese School, Edinburgh · Prepared lessons and taught mathematics.

Publications

- Zhao, P.**, Sansone, E., Kryven, M., & Zhao, B. (2026). Online library learning in human visual puzzle solving. *Proceedings of the 48th Annual Meeting of the Cognitive Science Society*. [Proceedings](#)
- Bao, Q., Yang, F., & **Zhao, P.** (2025). Exploring generative AI empowerment in city brand building from a human-factors perspective: An empirical study based on Wuhan city emoji. *Human Factors in Communication of Design, AHFE Conference Proceedings*. [DOI](#)

Preprints

- Hernandez Cano, L., Zareski, I.*, El Amouri, L.*, **Zhao, P.***, Mascini, M., Sansone, E., Pu, Y., Zhao, B., & Kryven, M. (2026). Prospective compression in human abstraction learning. *arXiv preprint*, arXiv:2605.09985. Under review. [arXiv](#)

* Equal contribution.

Conference Abstracts

- Zareski, I., El Amouri, L., **Zhao, P.**, Hernandez Cano, L., Sansone, E., Zhao, B., & Kryven, M. (2026). Adaptive online learning of structured abstractions in visual problem-solving. *Cognitive Computational Neuroscience (CCN 2026)*. Extended abstract; poster A58. [Conference record](#)

Talks and Presentations

- Aug 2026 **Human online library learning**
Lightning talk · IICCSSS, Tübingen, Germany
- Jul 2026 **Online library learning in human visual puzzle solving**
Lightning talk (virtual) · CogSci 2026, Rio de Janeiro, Brazil
- May 2026 **Prospective compression in human abstraction learning**
CogSciCon, Edinburgh, UK
- Jul 2025 **Conference Presenter** · AHFE International Conference
Presented the co-authored study on generative AI and Wuhan city emoji.
- Jan 2025 **Online library learning in human visual puzzle solving**
Lab talk · Bramley Lab, University of Edinburgh, UK

Summer school

- Aug 2026 **International Interdisciplinary Computational Cognitive Science Summer School (IICCSSS)**, Tübingen, Germany
- Jul 2026 **Southampton Summer Statistics School**
University of Southampton · Online
- Jul–Aug 2024 **Peking University Summer School**
Coursework in neuroaesthetics, including critical analysis of fMRI and TMS studies.
- Jun–Jul 2024 **Shanghai Jiao Tong University Summer School**
Coursework on sociocultural perspectives on inequality.

Research Skills

- Programming** Python, R, MATLAB, JavaScript, HTML/CSS, SQL.
- Computational cognition** Program synthesis, library learning, model comparison.
- Machine learning** Supervised learning, predictive modelling.
- Behavioural research** Experimental design, statistical analysis of behavioural data.
- Languages** Chinese (native), English (proficient).